

# MTH601 By Sparrow

## Final Term

Fall 2021-2022

1. When money value changes with time at 10 %, then PWF for the first year is  
A. 1  
B. 0.909  
C. 0.852  
D. 0.9
2. When money value changes with time at 20%, the discount factor for the 2nd year is  
A. 1  
B. 0.833  
C. 0  
D. 0.6955
3. Which of the following replacement policies is considered to be dynamic in nature?  
A. Time is a continuous variable and the money value does not change with time  
B. When money value does not change with time and time is a discrete variable  
C. When money value changes with time  
D. When money value remains constant for some time and then goes on changing with time
4. When the probability of failure reduces gradually, the failure mode is said to be  
A. Regressive  
B. Retrogressive  
C. Progressive  
D. Recursive
5. \_\_\_\_\_ is concerned with the determination of the most economic replacement policy.  
A. Probabilistic programming

- B. Linear programming
  - C. Search theory
  - D. Replacement theory**
6. Which cost of the following is irrelevant to replacement analysis?
- A. Purchase cost of the machine
  - B. Operating cost of the machine
  - C. Maintenance cost of the machine
  - D. Machine hour rate of the machine**
7. If a balanced Transportation problem with '7' sources and '6' sinks then its non-degenerate basic feasible solution will have----- independent allocations.
- A. 10**
8. Group replacement policy is most suitable for
- A. Trucks
  - B. Infant machines
  - C. Street light bulbs**
  - D. New cars
9. Replacement of an item will become necessary when
- A. an old item becomes too expensive to operate or maintain**
  - B. when your operator desires to work on a new machine
  - C. when your opponent changes his machine in his unit
  - D. when the company has surplus funds to spend
10. The following is not discussed in group replacement policy
- A. Failure Probability
  - B. Cost of individual replacement
  - C. Loss due to failure
  - D. Present worth factor series**
11. It is assumed that maintenance cost mostly depends on
- A. calendar age
  - B. manufacturing date
  - C. running age**
  - D. user's age
12. Replacement decision is very much common in this stage
- A. Infant stage

- B. Old age
  - C. Youth
  - D. In all the above**
13. The replacement policy that is imposed on an item irrespective of its failure is
- A. Group replacement**
  - B. Individual replacement
  - C. Repair spare replacement
  - D. Successive replacement
14. The total number of allocations in a basic feasible solution to the transportation problem of size is equal to
- A.  $M+N-1$**
15. Contractual maintenance or agreement maintenance with the manufacturer is suitable for equipment, which is
- A. In its infant state**
  - B. When a machine is an old one
  - C. Scrapped
  - D. None of these
16. Which of the following maintenance policy is not used in the old age stage of a machine?
- A. Operate up to failure and do corrective maintenance
  - B. Reconditioning
  - C. Replacement
  - D. Scheduled preventive maintenance**
17. The following replacement model is said to be a probabilistic model
- A. When money value does not change with time and time is a continuous variable
  - B. When money value changes with time
  - C. When money value does not change with time and time is a discrete variable
  - D. Preventive maintenance policy**
18. A machine is replaced with an average running cost
- A. Is not equal to the current running cost
  - B. till the current period is greater than that of next period
  - C. of the current period is less than that of next period**
19. Decreasing failure rate is usually observed in \_\_\_\_\_ stage of the machine

- A. **infant**
  - B. youth
  - C. old age
  - D. any time in its life
- 20. The type of failure that usually occurs in old age of the machine is
  - A. Random failure
  - B. Early failure
  - C. Chance failure
  - D. **Wear - out failure**
- 21. For the project of a firm, if '5' sales persons are assigned '5' different sales territories, then in how many ways a single territory can be assigned to a single sale person?
  - A. **120**
- 22. The production manager will not recommend group replacement policy in case of
  - A. When large number of identical items is to be replaced
  - B. Low cost items are to be replaced, where record keeping is a problem
  - C. For items that fail completely
  - D. **Repairable items**
- 23. In replacement analysis the maintenance cost is a function of
  - A. **time**
  - B. resale value
  - C. initial investment
  - D. none of these
- 24. Which of the following is the correct assumption for replacement policy when money value does not change with time?
  - A. No Capital cost
  - B. No scrap value
  - C. **Constant scrap value**
  - D. Zero maintenance cost
- 25. In the assignment problem, the cost matrix is \_\_\_\_\_.
  - E. **rectangular**
- 26. Which one of the following does not match the group?

- A. Present Worth Factor (PWF)
  - B. Discounted rate (DR)
  - C. Depreciation value (DV)
  - D. Mortality Tables (MT)**
27. Group replacement policy applies to
- A. Irreparable items
  - B. Repairable items
  - C. Items that fail partially
  - D. Items that fail completely & suddenly**
28. If a machine becomes old, then the failure rate expected will be
- A. Constant
  - B. Increasing**
  - C. decreasing
  - D. we cannot say
29. Replacement is said to be necessary if
- A. Failure rate is increasing
  - B. Failure cost is increasing
  - C. Failure probability is increasing
  - D. Any of these**
30. In this stage, the machine operates at highest efficiency and its production rate will be high
- A. Infant stage
  - B. Youth stage**
  - C. Old age
  - D. None of these
31. In retrogressive failures, the failure probability \_\_\_\_\_ with time
- A. increases
  - B. remains constant
  - C. decreases**
  - D. none of these
32. The objective of a transportation problem is to develop an \_\_\_\_\_ transportation schedule that meets all demand from given stock at a \_\_\_\_\_ total shipping cost.
- A. (integral, maximum)**
33. For North West Corner method, in the first row and first column, resource and sink

contain '5' and '7' units respectively; then after allocating the appropriate amount 'x11' in the cell (1,1), we will move towards which of the following cell?

A. (2,2)

34. While solving an LP problem by the Simplex method, in the standard table, the element at the intersection of key column and key row is called ----- element.

A. Pivot

35. Search and cutting methods are used to solve following types of problems

A. Transportation

B. Assignment

B. Integer Programming

C. Queuing problems

36. To identify and maintain the proper precedence relationship between activities those are not connected by event, we introduce

A. Parallel Activity

B. Dummy Activity

C. Sequential Activity

D. None of these

37. The positive slack variables indicate \_\_\_\_\_.

A. excess resource available (sure)

38. The variable is chosen by examining the cost coefficients in the objective function.

A. Entering

B. Leaving

39. In retrogressive failures, the failure probability \_\_\_\_\_ with time

A. Increases

B. Remains

C. Constant

D. Decreases

40. Decreasing failure rate is usually observed in \_\_\_\_\_ stage of the machine

A. Infant

B. Youth

41. When money value changes with time at 10 %, then PWF for first year is

A. 1

- B. 0.909**  
C. 0.852
42. The rent for the stores where materials are stored falls under  
**A. Inventory carrying cost**  
B. Ordering cost
43. O.R. Society of India is founded in the year  
A. 1965  
B. 1970  
**C. 1959**
44. The rule which determines the decision at each stage is known as  
A. State  
B. Stage  
**C. Policy**  
D. Decision
45. In dynamic programming problems, the main problem is divided into sub problems.  
Each sub problem is known as  
A. Part  
**B. Stage**  
**C. State**
46. A competitive situation is known as  
A. Competition  
B. Marketing  
**C. Game**  
D. None of the above
47. In case, there is no saddle point in a game then the game is  
A. Deterministic game  
B. Fair game  
**C. Mixed strategy game**  
D. Multi player game
48. When Minimax and Maximum criteria matches, then  
A. Fair game is exists  
B. Unfair game is exists  
C. Mixed strategy exists  
**D. Saddle point exists**

49. If the value of the game is zero, then the game is known as

- A. Fair strategy
- B. Pure strategy
- C. Pure game**
- D. Mixed strategy

50. The objective of network analysis is to\_\_\_\_\_.

- A. minimize total project duration**
- B. minimize total project cost
- C. minimize production delays, interruption and conflicts
- D. maximize total project duration

51. Service mechanism in a queuing system is characterized by\_\_\_\_\_.

- A. customers behavior
- B. servers behavior**
- C. customers in the system
- D. server in the system

52. The difference between total float and head event slack is \_\_\_\_\_

- A. free float**
- B. independent float
- C. interference float
- D. linear float

53. For any primal problem and its dual\_\_\_\_\_.

- A. optimal value of objective function is same
- B. dual will have an optimal solution iff primal does too
- C. primal will have an optimal solution iff dual does too**
- D. both primal and dual cannot be infeasible

54. If any value in XB column of final simplex table is negative, then the solution is \_\_\_\_\_.

- A. Infeasible
- B. Infeasible
- C. Bounded**
- D. no solution

55. Operations Research approach is \_\_\_\_\_.

- A. multi-disciplinary**

- B. scientific
  - C. intuitive
  - D. collect essential data
56. If there is no non-negative replacement ratio in solving a Linear Programming Problem then the solution is \_\_\_\_\_.
- A. Feasible
  - B. Bounded
  - C. Unbounded
  - D. Infinite
57. The calling population is considered to be infinite when \_\_\_\_\_.
- A. all customers arrive at once
  - B. capacity of the system is infinite
  - C. service rate is faster than arrival rate
  - D. arrivals are independent of each other
58. In marking assignments, which of the following should be preferred?
- A. Only row having single zero
  - B. Only column having single zero
  - C. Only row/column having single zero
  - D. Column having more than one zero
59. A petrol pump has one pump; Vehicles arrive at the petrol pump according to poison input process at average of 12 per hour. The service time follows exponential distribution with a mean of 4 minutes. The pumps are expected to be idle for \_\_\_\_\_.
- A. 3/5
  - B. 4/5
  - C. 5/3
  - D. 6/5
60. If the order quantity (size of order) is increased, \_\_\_\_\_
- A. holding costs decrease and ordering costs increase
  - B. holding costs increase and ordering costs decrease
  - C. the total costs increase and then decrease
  - D. storage cost as well as stock-out cost increase
61. A mixed strategy game can be solved by \_\_\_\_\_.
- A. Simplex method

- B. Hungarian method
  - C. Graphical method**
  - D. Degeneracy
62. The activity cost corresponding to the crash time is called the \_\_\_\_\_.
- A. critical time
  - B. normal time
  - C. cost slope
  - D. crash cost**
63. A set of feasible solution to a Linear Programming Problem is \_\_\_\_\_
- A. convex**
  - B. polygon
  - C. triangle
  - D. bold
64. If the primal problem has  $n$  constraints and  $m$  variables then the number of constraints in the dual problem is \_\_\_\_\_.
- A.  $mn$**
  - B.  $m+n$
65. The non-basic variables are called \_\_\_\_\_.
- A. shadow cost**
  - B. opportunity cost
  - C. slack variable
  - D. surplus variable
66. Key element is also known as \_\_\_\_\_.
- A. slack
  - B. surplus
  - C. artificial
  - D. pivot**
67. The allocation cells in the transportation table will be called \_\_\_\_\_ cell
- A. Occupied**
68. To resolve degeneracy at the initial solution, a very small quantity is allocated in \_\_\_\_\_ cell
- A. occupied
  - B. unoccupied**

69. An assignment problem is a particular case of \_\_\_\_\_.  
A. **transportation Problem**  
B. assignment Problem  
C. travelling salesman problem  
D. replacement Problem
70. Using \_\_\_\_\_ method, we can never have an unbounded solution  
A. Simplex  
B. **Dual simplex**
71. The customers of high priority are given service over the low priority customers is \_\_\_\_\_.  
A. **Pre-Emptive**
72. A queuing system is said to be a \_\_\_\_\_ when its operating characteristic are independent upon time  
A. pure birth model  
B. pure death model  
C. transient state  
D. **steady state**
73. The difference between total and free float is \_\_\_\_\_.  
A. total  
B. free  
C. independent  
D. **interference**
74. The number of time estimates involved in Program Evaluation Review Technique problem is \_\_\_\_\_.  
A. 1  
B. 2  
C. **3**  
D. 4
75. The assignment problem is always a \_\_\_\_\_ matrix.  
A. circle  
B. **square**  
C. rectangle  
D. triangle

76. The slack variables indicate \_\_\_\_\_.

- A. excess resource available.
- B. shortage of resource
- C. nil resource
- D. idle resource**

77. While solving a linear programming problem infeasibility may be removed by \_\_\_\_\_.

- A. adding another constraint
- B. adding another variable
- C. removing a constraint**
- D. removing a variable

78. The right hand side constant of a constraint in a primal problem appears in the corresponding dual as \_\_\_\_\_.

- A. a coefficient in the objective function**

79. During iteration while moving from one solution to the next, degeneracy may occur when \_\_\_\_\_.

- A. the closed path indicates a diagonal move
- B. Two or more occupied cells on the closed path with minus sign are tied for lowest circled value.**

80. Maximization assignment problem is transformed into a minimization problem by \_\_\_\_\_.

- A. adding each entry in a column from the maximum value in that column
- B. subtracting each entry in a column from the maximum value in that column
- C. subtracting each entry in the table from the maximum value in that table**
- D. adding each entry in the table from the maximum value in that table

81. Priority queue discipline may be classified as \_\_\_\_\_.

- A. pre-emptive or non-pre-emptive
- B. limited
- C. unlimited**
- D. finite

82. Replace an item when \_\_\_\_\_.

- A. average cost upto date is equal to the current maintenance cost**

- B. average cost upto date is greater than the current maintenance cost
  - C. Average cost upto date is less than the current maintenance cost.
  - D. next year running cost is more than average cost of nth year
83. In time cost trade off function analysis\_\_\_\_\_.
- A. cost decreases linearly as time increases
  - B. cost increases linearly as time decreases
  - C. cost at normal time is zero
  - D. cost increases linearly as time increases
84. The minimum number of lines covering all zeros in a reduced cost matrix of order n can be \_\_\_\_\_.
- A. at the most n
  - B. at the least n
  - C. n-1
  - D. n+1
85. For a 2.5% increase in order quantity (under fundamental EOQ problem) the total relevant cost would \_\_\_\_\_.
- A. Increase by 2.5%.
  - B. Decrease by 2.5%.
86. In the basic EOQ model, if the lead time increases from 2 to 4 days, the EOQ will \_\_\_\_\_.
- A. double increase
  - B. remain constant
  - C. but not double
  - D. decrease by a factor of two
87. In the network, one activity may connect any \_\_\_\_\_ nodes
- A. 1
  - B. 2
  - C. 3
  - D. 4
88. Graphical method is also known as \_\_\_\_\_.
- A. Simplex Method
  - B. Dual Simplex Method
  - C. Big-M Method
  - D. Search-Approach Method

89. Linear Programming Problem is a technique of finding the \_\_\_\_\_.

- A. optimal value
- B. approximate value
- C. initial value
- D. infeasible value

90. The irreducible minimum duration of the project is called \_\_\_\_\_.

- A. critical time
- B. normal time
- C. cost slope
- D. crash duration

91. The cost of a slack variable is \_\_\_\_\_.

- A. 0
- B. 1
- C. 2
- D. -1

92. The server utilization factor is also known as \_\_\_\_\_

- A. exponential distribution
- B. traffic intensity

93. A Linear Programming Problem have \_\_\_\_\_ optimal solution

- A. 1
- B. 2
- C. more than 1

94. The models in which only arrivals are counted and no departure takes place are called \_\_\_\_\_.

- A. pure birth model
- B. pure death mode

95. \_\_\_\_\_ of a queuing system is the state where the probability of the number of customers in the system depends upon time

- A. pure birth model
- B. pure death model
- C. transient state
- D. steady state

96. The initial event which has all outgoing arrows with no incoming arrow is numbered \_\_\_\_\_.  
A. 0  
B. 1  
C. -1  
D. 2
97. In a network diagram an event is denoted by the symbol \_\_\_\_\_.  
A. arrow  
B. straight line  
C. curve  
D. circle
98. \_\_\_\_\_ is used for non-repetitive jobs  
A. Queue  
B. Replacement  
C. CPM  
D. PERT
99. An \_\_\_\_\_ represent the start or completion of some activity and as such it consumes no time  
A. activity  
B. event  
C. slack  
D. path
100. The region common to all the constraints including the non-negativity restrictions is called the \_\_\_\_\_.  
A. solution space  
B. unique solution  
C. optimum solution  
D. infeasible solution
101. Operation research approach is typically based on the use of \_\_\_\_\_.  
A. Physical model.  
B. Mathematical model.  
C. Iconic model.  
D. Descriptive model.
102. For a salesman who has to visit  $n$  cities, following are the ways of his tour plan \_\_\_\_\_.

- A.  $n!$
  - B.  $(n+a)!$
  - C.  $(n-a)!$
  - D.  $N$
103. The problem of replacement is felt when job performing units fail \_\_\_\_\_.
- A. suddenly and gradually
  - B. gradually
  - C. suddenly
  - D. neither gradually nor suddenly
104. Float analysis is useful for \_\_\_\_\_.
- A. total float
  - B. free float
  - C. independent float
  - D. variance of each float
105. The difference between free float and tail event slack is \_\_\_\_\_.
- A. total float
  - B. independent float
  - C. interference float
  - D. slack
106. The average arrival rate in a single server queuing system is 10 customers per hour and average service rate is 15 customers per hour. The average time that a customer must wait before it is taken up for service shall be \_\_\_\_\_ minutes.
- A. 6
  - B. 8
  - C. 10
  - D. 12
107. The time between the placement of an order and its delivery is called as \_\_\_\_\_.
- A. buffer time
  - B. lead time
  - C. Economic Order Quantity
  - D. capital time

108. All the basis for a transportation problem is \_\_\_\_\_.  
A. square  
B. rectangle  
C. diagonal  
**D. triangle**
109. The linear function to be maximized or minimized is called \_\_\_\_\_ function.  
A. injective  
B. surjective  
C. bijective  
**D. optimal**
110. The coefficient of an artificial variable in the objective function of penalty method is always assumed to be \_\_\_\_\_.  
A. 0  
B. 1  
C. M  
**D. -M**
111. The process that performs the services to the customer is known as \_\_\_\_\_.  
A. queue  
**B. service channel**  
C. customers  
D. server
112. A queuing system is said to be a \_\_\_\_\_ when its operating characteristic are dependent upon time  
A. pure birth model  
B. pure death model  
**C. transient state**  
D. steady state
113. Slack is also known as \_\_\_\_\_.  
**A. Float**  
B. event  
C. activity

- D. path
114. What type of distribution does a time follow in program evaluation review technique model?
- A. Poisson
  - B. Exponential
  - C. Normal
  - D. Chi Square
115. A activity in a network diagram is said to be \_\_\_\_\_ if the delay in its start will further delay the project completion time.
- A. Critical
  - B. critical path
  - C. crash
  - D. non critical
116. In the optimal simplex table,  $Z_j - C_j = 0$  value indicates \_\_\_\_\_.
- A. alternative solution
  - B. bounded solution
  - C. infeasible solution
  - D. unbounded solution
117. In an assignment problem involving 5 workers and 5 jobs, total number of assignments possible are \_\_\_\_\_.
- A. 5
  - B. 10
  - C. 15
  - D. 20
118. Average number of trains spent in the yard is denoted by \_\_\_\_\_.
- A.  $E(n)$
  - B.  $E(m)$
  - C.  $E(v)$
  - D.  $E(w)$
119. The dual of the dual is \_\_\_\_\_.
- A. dual-primal
  - B. primal-dual
  - C. dual
  - D. primal

120. All equality constraints can be replaced equivalently by \_\_\_\_\_ inequalities
- A. 1
  - B. 2**
  - C. 3
  - D. 4
121. If the primal has an unbound objective function value then the dual has \_\_\_\_\_.
- A. solution
  - B. basic solution
  - C. basic feasible solution
  - D. no feasible solution**
122. An activity is represented by a/an \_\_\_\_\_.
- A. arrow**
  - B. straight line
  - C. curve
  - D. arc
123. The penalty for not taking correct decision is known as \_\_\_\_\_.
- A. fine
  - B. loss
  - C. cost
  - D. opportunity cost**
124. Every basic feasible solution of a general assignment problem having a square pay-off matrix of order  $n$  should have assignments equal to \_\_\_\_\_.
- A.  $2n-1$**
  - B.  $n$
  - C.  $n+1$
  - D.  $n-2$
125. A game is said to be fair if \_\_\_\_\_.
- A. lower and upper values are zero**
  - B. only lower value to be zero
  - C. only upper value to be zero
  - D. lower and upper values are not equal to zero

126. When  $D=18000$ , holding cost=Rs.1.20, set-up cost=Rs.400 ,EOQ = \_\_\_\_\_  
A. 3465  
B. 3750  
C. 3500  
D. 4000
127. Given arrival rate = 15/hr, service rate = 20/hr, the value of traffic intensity is \_\_\_\_\_.  
A.  $3/4$   
B.  $4/3$   
C.  $3/5$   
D.  $4/5$
128. A solution can be extracted from a model either by  
A. Conducting experiments on it  
B. Mathematical analysis  
C. Both A and B  
D. Diversified Techniques
129. An assignment problem can be viewed as a special case of transportation problem in which the capacity from each source is \_\_\_\_\_ and the demand at each destination is \_\_\_\_\_.  
A. 1; 1  
B. Infinity; infinity  
C. 0; 0  
D. 1000; 1000  
E. -1; -1
130. Hungarian Method is used to solve  
A. A transportation problem  
B. A travelling salesman problem
131. In Degenerate solution value of objective function \_\_\_\_\_.  
A. One or more basic variables are zero
132. In game theory, the outcome or consequence of a strategy is referred to as the

A. **Payoff.**

B. Penalty.

133. Operations research is the application of \_\_\_\_\_ methods to arrive at the optimal Solutions to the problems.

A. economical

B. **scientific**

C. a and b both

134. The degeneracy in the transportation problem indicates that

A. Dummy allocation(s) needs to be added

B. The problem has no feasible solution

C. **The multiple optimal solution exist**

135. The initial solution of a transportation problem can be obtained by applying any known method. However, the only condition is that

A. The solution be optimal

B. **The rim conditions are satisfied**

C. The solution not be degenerate

D. All of the above

136. The method of finding an initial solution based upon opportunity costs is called \_\_\_\_\_.

A. the northwest corner rule

B. **Vogel's approximation**

C. Johansson's theorem

137. The objective function and constraints are functions of two types of variables, \_\_\_\_\_ variables and \_\_\_\_\_ variables.

A. Positive and negative

B. **Controllable and uncontrollable**

138. The objective function for a minimization problem is given by

$$z = 2x_1 - 5x_2 + 3x_3$$

The hyperplane for the objective function cuts a bounded feasible region in the space

( $x_1$ ,  $x_2$ , and  $x_3$ ). Find the direction vector  $d$ , where a finite optimal solution can be reached.

A.  $d(2, -5, 3)$

**B.  $d(-2,5,-3)$**

C.  $d(2,5,3)$

139. What have been constructed from OR problems and methods for solving the models that are available in many cases?

A. Scientific Models

B. Algorithms

**C. Mathematical Models**

**Contact for Help.**

**We have checked all the Answers if you find any mistake let us know.**

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**Remember in Prayers.**

**Best Of Luck.**